

BIOLOGY

TAXONOMIC RANKS

Biologists classify organisms into groups according to their characteristics as a way of illustrating their evolutionary relationships. All life has descended from a single common ancestor, which lived billions of years ago, and has diversified through evolution. The most distantly related organisms belong to groups that are ranked as domains. These are subdivided into groups of successively lower ranks to contain organisms that are progressively more closely related. At any rank, modern

biologists strive to define groups that are monophyletic: ones that contain all organisms descended from a single point of ancestry. At the lowest rank, species, organisms are so closely related that they can usually interbreed. In the charts opposite and on pp.362–63, dotted lines are used to define informal assemblages of taxonomic ranks. Although these are not natural evolutionary groups, they often provide a convenient and useful way to refer to collections of organisms.

DOMAIN	KINGDOM	PHYLUM	CLASS	ORDER	FAMILY	GENUS	SPECIES
Introduced in the 1990s in response to discoveries that were made in cellular biology, the domain rank defines groups that represent the most ancient divisions of life on Earth. These emerged as distinct groups around 4 billion years ago and include bacteria and the more complex multicelled life forms.	Domains are divided into different kingdoms that include familiar groups of organisms (animals, plants, and fungi)—as well as other groups (including a range of single-celled organisms and algae) that evolved around a billion years ago.	Kingdoms are divided into phyla (singular: phylum). Organisms united in a phylum share a particular body plan; for animals and plants, this includes groups that originated 0.5–1 billion years ago, when the seas were teeming with life and land was being colonized for the first time.	Phyla are divided into classes. Organisms within a class are defined by body structure and life history. For example, land vertebrates are split into classes that include amphibians, reptiles, birds, and mammals. Plant classes include monocotyledon and dicotyledon flowering plants.	Classes are divided into orders. Animal orders are based largely on body structure, but plant orders are also defined by the chemicals made in their tissues. For example, the mammals are split into orders that include primates, rodents, and bats. Plant orders include Ranunculales (buttercups and relatives) and Lamiales (mints and relatives).	Orders are divided into families. Names of plant, algal, and fungal families conventionally end in “-aceae” (for example, Liliaceae for the lily family); names of animal families end in “-idae” (for example, Sciuridae for the squirrel family).	Families are divided into genera (singular: genus). The name of each genus is the first part of the scientific name of a species. For example, big cats are united in the genus <i>Panthera</i> and include the species <i>Panthera leo</i> (lion) and <i>Panthera tigris</i> (tiger). The generic and species names are always given in italics.	The species group is the only taxonomic rank that can be defined in biological terms. It is often taken to be a group of organisms that can interbreed. But in practice, most species—including those that cannot reproduce sexually or where reproductive biology is unknown—are defined by their physical characteristics.

DOMAINS AND KINGDOMS

Biologists used to divide all life into plants and animals, but the roots of biological diversity are now known to be far more complex. The earliest life diversified as single-celled organisms. Two lineages of these organisms, out of countless others, later gave rise to plants and animals. This means that most groups at the ranks of domain and kingdom include single-celled organisms. The most basic differences evolved when the earliest life split into three domains: simple single-celled Bacteria and Archaeae; and the more complex Eukaryotes, whose genetic material became packaged into a cellular nucleus. Many groups of eukaryotic organisms retained their single-celled nature, but others formed the multicelled bodies that became fungi, plants, and animals.

